

LAB 3 - Component Modelling & Architectural Pattern Selection

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Architecture Selection

We chose Layered Architecture for the Personal Subscription & Recurring Expense Auditor system. The architecture is organized into Presentation, Business, and Data layers, separating user interaction, application processing, and transaction data storage.

Architectural Choice

The Presentation Layer handles interaction with the Individual User and Finance Auditor. The Business Layer contains the Transaction Import, Expense Analysis, Subscription Management, Notification & Alert, and Audit & Reporting components. The Data Layer contains the Transaction Database. This structure reflects the major functions identified in the system requirements and use-case flow.

Reason 1 – Separation of Concerns

The system performs distinct activities such as importing CSV/OFX transaction data, identifying recurring billing patterns, managing subscriptions, generating renewal alerts, and producing audit reports. Layered Architecture separates these responsibilities into appropriate components, making the system easier to understand and maintain.

Reason 2 – Maintainability

The components can be changed within their respective responsibilities with less impact on other parts of the system. For example, the recurring-expense analysis logic can be modified without requiring changes to the presentation or database layer. This supports the system as its functionality evolves.

Security Advantage

Layered Architecture separates the presentation layer from transaction processing and data storage, helping restrict direct access to sensitive financial information. This supports the security requirement that financial transaction imports be processed securely without storing raw banking credentials in cleartext.

Performance Benefit

Separating presentation from business processing allows transaction analysis to be performed independently of the user interface. This supports the performance requirement to process a transaction file containing up to 10,000 transactions and display the analysis results within 5 seconds under normal operating conditions.